

Summary

The preservation of global biodiversity in expansive sanctuaries like Etosha National Park presents a critical conflict between immense geographical scale and limited human resources. To address this challenge, this paper proposes a spatial-time mathematical framework designed to optimize strategic resource deployment and quantify protection levels across both space and time. Our approach follows a rigorous, integrated logical chain that initiates with multidimensional Risk Mapping, progresses through Topological Decomposition and Geometric Routing, and culminates in Spatiotemporal Scheduling and Staffing Allocation.

At the foundational level, we construct a comprehensive Risk Evaluation Model that synthesizes human-driven threats and natural environmental volatility into a unified, actionable metric for prioritization, ensuring that limited assets are concentrated where they are most critically needed.

Based on the scenario above, we introduce a Topology Decomposition layer that partitions the park into discrete, traversable units based on connectivity and terrain constraints. By identifying feasible patrol domains and impassable regions, we convert the risk control problem of the complicated park into an organized set of lower-dimensional planning decisions, obtaining dispatchable methods that are effective.

Within these units, we make innovations based on the Active Contour (Snake) algorithm. It regulates path tension and curvature to account for maneuverability constraints and thus generate a smooth, closed patrol backbone. This spatial skeleton is further refined by Two-Anchor Outward-Bulging Petals that target residual high-risk clusters via an optimized Cost-Benefit Selection process. This iterative expansion ensures that the final route remains closed and highly executable while maximizing risk-weighted protection with minimal marginal detour costs.

The technical core of our work lies in the Kalman-Nyquist Synthesis for patrol scheduling, which redefines wildlife protection as a Signal Reconstruction and Uncertainty Bounding problem. We employ the Nyquist-Shannon Sampling Theorem to establish a macro-level frequency baseline that prevents the temporal aliasing of rapidly shifting risk signals. Beyond simple reconstruction, we derive a closed-form solution for the Safety-Critical Revisit Time Interval based on the steady-state of the Kalman Filter Riccati equation. This time skeleton guarantees that the system's estimation uncertainty never exceeds a critical safety threshold.

This spatial-time scheduling allows us to translate abstract threat dynamics into mechanistically derived staffing requirements, calculate the minimum manpower needed to suppress the growth of spatiotemporal entropy across the park, and validate the robustness of this strategy through Dynamic Control simulations and the Fourier Amplitude Sensitivity Test (FAST), demonstrate the global scalability of our framework by successfully adapting it to Barro Colorado Nature Monument and Bavarian Forest National Park, and provide a sustainable and quantifiable blueprint for global wildlife conservation.

Keywords: National Park Protection Management, 0D Persistent Homology, Active Contour, Orienteering, Nyquist-Shannon Sampling Theorem, Kalman Filter, FAST

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1 Introduction

1.1 Background

The preservation of global biodiversity stands as one of the most profound challenges of the 21st century, representing a collective commitment to safeguarding the planet's irreplaceable ecological heritage. National parks and wildlife reserves serve as the final bastions for endangered species, yet their immense geographical scale often creates a critical vulnerability: the vastness required for species to thrive simultaneously renders continuous protection logistically unattainable. Etosha National Park in Namibia, a sprawling sanctuary of over 22,000 km², epitomizes this global struggle, where a limited force of fewer than 300 personnel must defend against the escalating threats of illegal poaching, habitat disruption, and environmental volatility. In this context, the integration of human vigilance and advanced technological assets is not merely a logistical choice, but a strategic imperative. By developing a robust mathematical framework to optimize resource allocation and quantify protection, we aim to provide a scalable and sustainable blueprint that ensures the long-term survival of wildlife in an increasingly resource-constrained world.



Figure 1: Wildlife Protection

1.2 Problem Restatement

To address the complex challenges of wildlife conservation at scale, our team has broken down the problem into specific objectives and a sequential modeling process.

- **Objective 1:** Define and quantify wildlife protection metrics.
- **Objective 2:** Prioritize conservation threats and species risks.

- **Objective 3:** Optimize strategic allocation of limited resources.
- **Objective 4:** Estimate staffing needs for long-term protection.
- **Objective 5:** Generalize the model to global protected areas.

With the objectives identified, a specific breakdown of tasks is also provided below.

- **Task 1: Defining Protection.** Identify major risks and establish a quantitative metric to measure wildlife protection levels across the park.
- **Task 2: Resource Allocation.** Develop a mathematical model to strategically deploy personnel and technology under limited resource constraints.
- **Task 3: Staffing Analysis.** Evaluate protection effectiveness over time and estimate the human personnel required for sustainable management.
- **Task 4: Sensitivity Analysis.** Sensitivity Analysis Investigate how changes in assets, personnel, or schedules affect the overall protection strategy and security.
- **Task 5: Model Adaptation.** Demonstrate the model's flexibility by adapting it to two other national parks on different continents with modified parameters.

1.3 Our Work

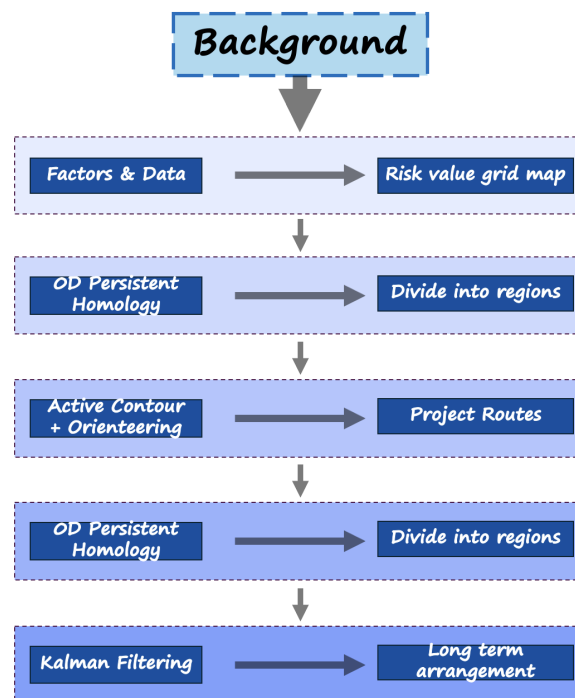


Figure 2: Model Overview

2 Variables and Parameters

Table 1: Description of Major Variables and Parameters

Variable	Description	Unit	Variable	Description	Unit
$r_{ij}^{(P)}$	Poaching risk index at grid (i, j)	-	$D_{ij}^{(b)}$	Distance to the nearest boundary	km
$A_{ij,s}$	Conservation significance of species s	-	$r_{ij}^{(C)}$	Human-wildlife conflict risk index	-
$D_{ij}^{(r)}$	Distance to the nearest road	km	$r_{ij}^{(L)}$	Illegal logging probability	-
V_{ij}	Vegetation height at grid (i, j)	m	$r_{ij}^{(d)}$	Natural disaster risk index	-
t_0, t_d	Current time and event occurrence time	days	$\mathbf{R}_u$	Total relative risk value for grid u	-
R	Spatial risk map (threat intensity)	-	M	Binary traversability mask (0 or 1)	-
d	Distance to nearest impassable cell	km	U	Risk benefit attraction field	-
O	Obstacle potential repulsion field	-	p_i, v_i	Active contour control points	-
F_i	Total force acting on control point i	-	α, β	Elasticity and smoothness weights	-
γ, μ	Risk attraction and obstacle repulsion	-	P_{main}	Discretized main patrol backbone	-
H	Residual risk demand field	-	C_j	Spatial residual high-risk cluster j	-
M_j	Residual mass of cluster j	-	c_j	Weighted centroid of cluster j	-
r_j	Equivalent radius of cluster j	km	A_j, B_j	Anchor coordinates for petals	-
ΔL_j	Net detour length of the petal	km	G_j	Marginal protection gain of petal	-
$S(P)$	Global Protection Score of route P	-	τ_k	Patrol revisit interval for unit k	days
f_k^*	Implemented patrol frequency	day^{-1}	P_{max}	Safety threshold for uncertainty	-
Q_c	Process noise (threat emergence rate)	day^{-1}	R_{obs}	Observation noise variance	-

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Variable	Description	Unit	Variable	Description	Unit
L_k	Length of patrol route $\mathcal{L}_k$	km	v_{avg}	Average patrol movement speed	km/h
H_{max}	Maximum working hours per shift	hours	N	Total personnel required	persons
m_0	Number of rangers per team	persons	T_k	Cycle workload for a specific route	hours
$\omega_{\text{max},k}$	Highest significant threat frequency	day^{-1}	τ_{Kalman}	Safety-Critical Revisit Interval	days

3 Assumptions and Justifications

- **Assumption 1: The park can be discretized into a grid of cells.**

Justification: Etosha's defining geographical feature is its aridity. Unlike dense tropical forests where wildlife is evenly distributed, Etosha's megafauna are tethered to water sources, especially during the dry season. This creates highly concentrated, predictable nodes of activity.

- **Assumption 2: Outposts can be built anywhere, regardless of the terrain, without harming the animal habitat.**

Justification: Modern technologies are able to remove or flat any obstacles that are in the way of outposts. The area an outpost take up is relatively small, which means it won't hurt the animal habitat a lot. Static sensor placements, such as monitory cameras, is usually unnoticeable to animals, which further decreases the harm of building artificial equipment in animal habitats.

- **Assumption 3: The probability of high-value animal presence is negatively correlated with the distance to the nearest active water source.**

Justification: Etosha is an extreme, arid savanna ecosystem where surface water is the primary limiting factor for life. While animals disperse slightly during the brief wet season, for the majority of the year, megafauna (e.g. Black Rhino) are biologically tethered to natural and artificial waterholes. This predictability allows the model to treat waterholes as high-density "nodes" for both wildlife protection and poaching risk.

- **Assumption 4: Migrating birds always return to the same spot each year.**

Justification: Birds are exceptionally good at remembering the path they take in migration. In order to return to their specific reproduction points each year, they must have the ability to return to the same zone, allowing for localized and seasonal protection protocols.

- **Assumption 5: The potential harm resulting from poaching decreases exponentially as the distance form the edge of the park increases linearly.**

Justification: This assumption is grounded in the logistical constraints and cumulative risks faced by poachers. As a poacher travels linearly deeper into the park on foot, their logistical burden increases. More importantly, the time spent inside the reserve increases linearly, which exponentially increases their cumulative probability of being detected by ranger patrols or fixed sensors. Additionally, a significant portion of poaching is opportunistic subsistence hunting, which is strictly confined to the park perimeter. Therefore, modeling the risk—and the resulting expected harm—as an exponential decay function provides a mathematically rigorous way to prioritize boundary defenses.

4 Risk Evaluation Model

A risk evaluation model is established to quantify conservation challenges.

4.1 Model Overview

To establish a rigorous resource allocation, we identified the main conservation challenges faced by the park and categorized them into human-driven and natural threats, as shown in Figure 3. Each indicator is quantified by seasonal datasets, and they are combined, and the total relative risk value for a grid point is gained through the TOPSIS method.

4.2 Factors



Figure 3: Factors of Human-Driven and Natural Threats

4.3 Human-Driven Threats

These threats arise from humans' interference within the park.

4.3.1 Poaching

This poaching index describes the probability of poaching at a grid, which depends on its proximity to boundaries, historical poaching frequency, and high-value animal activity. The weights a are determined by AHP as $a_1 = 100$ and $a_2 = 20$.

$$r_{ij}^{(P)} = a_1 e^{-D_{ij}^{(b)}} + a_2 \sum_s A_{ij,s}. \quad (1)$$

In which $D_{ij}^{(b)}$ represents the minimum Euclidean distance between grid and the nearest boundary

$$D_{ij}^{(b)} = \min \left(\sum_{(m,n) \in \mathcal{P}(ij) \rightarrow b} \sqrt{(m-i)^2 + (n-j)^2} \right) \quad (2)$$

where s represents all wildlife that exists at the grid cell and α_s denotes the conservation significance of a species:

$$A_s = \begin{cases} 1, & s = \text{Endangered} \\ 0.3, & s = \text{Present} \\ 0.1, & s = \text{Abundant} \end{cases} \quad (3)$$

4.3.2 Human-Wildlife Conflict

This is quantified as the product of shortest distance from a grid cell to the nearest pedestrian road and an active species type at that site, given by

$$r_{ij}^{(C)} = e^{-D_{ij}^{(r)}} \sum_s A_{ij,s}. \quad (4)$$

In which $D_{ij}^{(r)}$ represents the minimum Euclidean distance between the grid and the nearest boundary, calculated through the same method as $D_{ij}^{(b)}$.

4.3.3 Illegal Logging

Illegal logging probability is determined by tree cover density across a 3×3 grid cluster, which δ_{ij} evaluates the existence of trees at a grid and V_{ij} represents vegetation height.

$$r_{ij}^{(L)} = \sum_{m=i-1}^{i+1} \sum_{n=j-1}^{j+1} \delta_{m,n} \quad (5)$$

$$\delta_{ij} = \begin{cases} 1, & V_{ij} > 5 \\ 0, & V_{ij} \leq 5 \end{cases} \quad (6)$$

4.4 Natural Threats

These threats arise from natural climates and topography.

4.4.1 Natural Disaster

The influences of natural disasters are divided into two parts: long-term frequency and short-term frequency. Long-term frequency refers to the frequency at which disaster occurs over an extended period, reflecting the long-term periodicity; short-term frequency refers to the occurrence frequency within a recent period, reflecting its recent probability. For instance, if a flood has already persisted for two days, the risk of it continuing to occur on the next day would be higher.

The total risk index of natural disaster at a given point is given by:

$$r_{ij}^{(d)} = \sum_{t_d < t_0} 0.99^{-(t_0 - t_d)} + \sum_{t=0}^{\Delta t} 1.2^t \quad (7)$$

The first term represents long-term frequency, in which t_0 represents the current time and t_d represents the times when poaching happened at the location, in days. This model shows an exponentially decaying relationship between the time elapsed since a poaching occurred and its influence on the risk evaluation of that location, which gives more emphasis on nearly happened poaching. The second term represents short-term frequency, where Δt represents the time elapsed from the first day of a disaster to the current time.

4.5 Ecological Vulnerability

Ecological vulnerability is determined by the number of species and vegetation abundance. Each grid cell's value depends on a surrounding 3×3 grid cluster:

$$r_{ij}^{(E)} = \sum_{m=i-1}^{i+1} \sum_{n=j-1}^{j+1} \left(\sum_s \alpha_s A_{mn,s} \cdot \frac{1}{V_{mn} + 10} \right). \quad (8)$$

Where the first term $\sum_s \alpha_s A_{mn,s}$ represents the number of values of species, while the second term quantifies the vegetation density. This equation implies that ecological vulnerability is high in areas with a high number of species and relatively sparse vegetation cover.

4.6 Total Relative Risk Value

To combine the five risk factors into a total representative value, TOPSIS is employed, which identifies the relative risk by calculating the Euclidean distance between each grid cell and the theoretical best and worst scenario.

In addition, in order to eliminate dimensional differences between indices, normalization is performed:

$$r_{ij}^* = \frac{r_{ij}}{\sqrt{\sum_{i=1}^n r_{ij}^2}} \quad (9)$$

Let n be the total number of grid cells and $m = 5$ be the number of risk indicators. The resulting normalized matrix is

$$\mathbf{R}^* = \left(r_{ij}^* \right)_{n \times m} \quad (10)$$

For each grid cell u , the total relative risk value $\mathbf{R}_u$ is determined by:

$$D_u^+ = \sqrt{\frac{1}{5} \sum_{k=1}^m \left(z_{uk} - z_k^+ \right)^2}, \quad D_u^- = \sqrt{\frac{1}{5} \sum_{k=1}^m \left(z_{uk} - z_k^- \right)^2} \quad (11)$$

where

$$\mathbf{R}_u = \frac{D_u^-}{D_u^+ + D_u^-} \quad (12)$$

Where z_k^+ and z_k^- represent the maximum and minimum value of an index, respectively, D_u^+ , D_u^- represent a grid's Euclidean distance to the extreme values.

4.7 Model Results

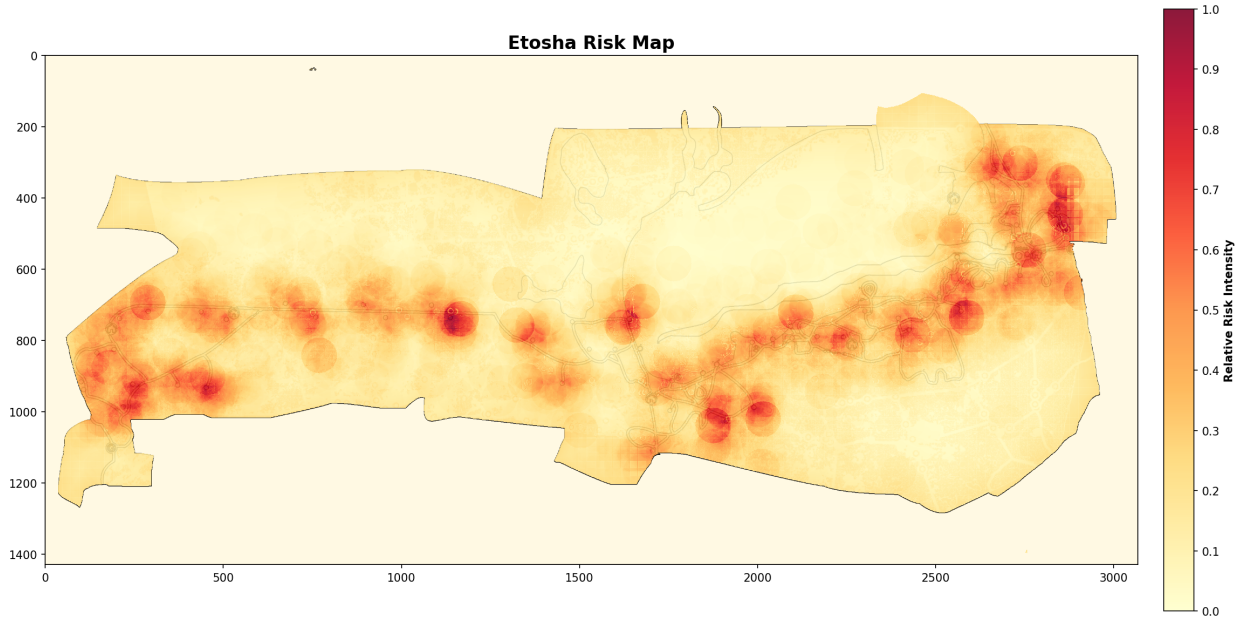


Figure 4: Risk Map of the Etosha National Park

5 Topological Decomposition of the Park

We introduce a topology-aware decomposition layer that bridges spatial risk representation and executable patrol planning.

Topological decomposition

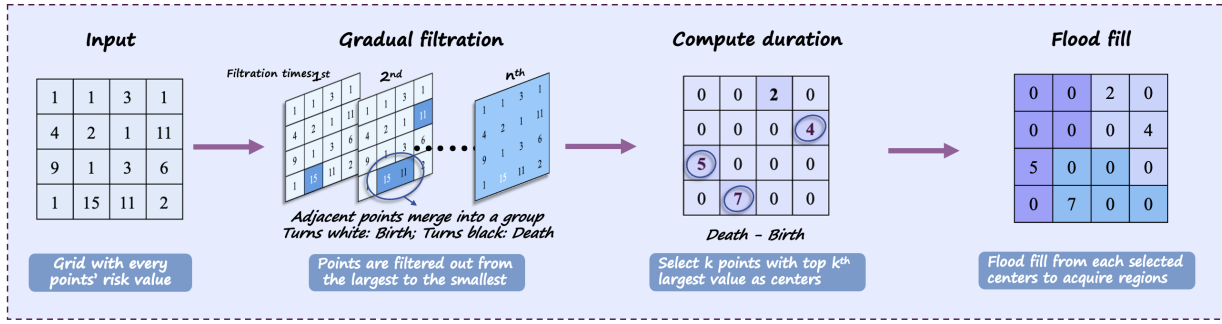


Figure 5: Topological Decomposition of the Park

5.1 Traversability and Connectivity Structure

To eliminate the reliance on subjective terrain-cost assumptions, we represent the environment through two fundamental inputs: the **Risk Map** $R(x, y) \geq 0$, which describes the spatial distribution of threat intensity, and the **Binary Traversability Mask** $M(x, y) \in \{0, 1\}$, where $M(x, y) = 0$ denotes impassable areas such as deep salt pans, steep slopes, or fenced boundaries. We then restrict the risk field to the feasible patrol domain by

$$\hat{R}(x, y) \leftarrow \hat{R}(x, y) \cdot M(x, y). \quad (13)$$

Impassable regions do not merely block motion locally. They create boundaries, bottlenecks, and corridor effects that shape the connectivity of the patrol space. To characterize this structure, we compute the Euclidean distance $d(x, y)$ from each passable cell to the nearest impassable cell $M = 0$. Cells near obstacle boundaries or narrow passages are more spatially constrained, whereas cells in open regions admit greater routing flexibility.

5.2 Topological Decomposition for Patrol Simplification

This structural layer **bridges risk evaluation and patrol deployment**. Without it, the patrol problem would remain a globally coupled optimization over a large continuous space with heterogeneous accessibility. By identifying the traversable domain and its connectivity constraints first, we convert the original park-scale problem into a more organized set of lower-dimensional planning decisions.

This decomposition **provides the basis for the subsequent routing model**: patrol backbones are generated within feasible connected regions, local refinements are added only where residual threats remain, and later patrol frequencies and manpower requirements can be assigned at the regional level. In this way, topological decomposition turns a large-scale protection problem into an executable deployment framework.

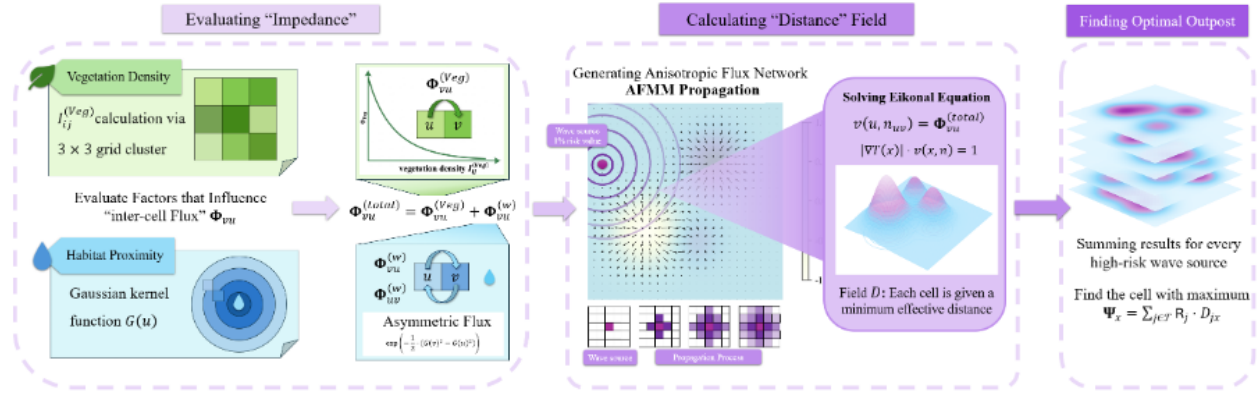


Figure 6: Establishing Outpost

After dividing the park into subareas, each of them is assigned an outpost to ensure the safety of wildlife. The site selection for outposts must balance multiple considerations: ensuring that patrol teams can rapidly approach high-risk areas while also preventing disturbance to wildlife habitat. In other words, outposts must maintain an appropriate distance from regions with high species density, such as waterholes. To capture this trade-off, we introduce the inter-cell flow to quantify the “impedance” exerted by water sources and dense vegetation during directional movement across the grid.

First, the fractional vegetation cover at grid cell (i, j) , denoted by $I_{ij}^{(\text{Veg})}$, is evaluated through the vegetation density over a 3×3 grid cluster, in which η_{ij} indicates the existence of vegetation.

$$I_{ij}^{(\text{Veg})} = \sum_{m=i-1}^{i+1} \sum_{n=j-1}^{j+1} \eta_{m,n} \quad (14)$$

$$\eta_{ij} = \begin{cases} 1, & V_{ij} > 0, \\ 0, & V_{ij} = 0. \end{cases} \quad (15)$$

One part of the inter-cell flow from grid cell u to v , denoted by $\Phi_{vu}^{(\text{Veg})}$, is modeled as an exponential decay function of vegetation density, reflecting the increasing impedance exerted by denser vegetation.

$$\Phi_{vu}^{(\text{Veg})} = \exp\left(-\frac{1}{9} \cdot \frac{I_u^{(\text{Veg})} + I_v^{(\text{Veg})}}{2}\right) \quad (16)$$

Another critical factor influencing inter-cell flow is the proximity to water sources, which represent important wildlife habitats. Empirical evidence suggests that human disturbance to wildlife habitats becomes negligible beyond a distance of 1000 m. To quantify this effect, we employ a Gaussian kernel with $\sigma = 500$ m, where $d(u, w_m)$ denotes the Euclidean distance from cell u to the m -th waterhole. The cumulative influence of all 86 waterholes on a grid cell is defined as

$$G(u) = \sum_{m=1}^{86} \exp\left(-\frac{d(u, w_m)^2}{2\sigma^2}\right) \quad (17)$$

Another part of the inter-cell flow is determined by the squared difference in the Gaussian potential values, which creates a directional flux with a negative gradient when approaching water areas. If the flow moves toward water, that is, when $G(v) > G(u)$, the exponent becomes negative, significantly reducing the flux $\Phi_{vu}^{(w)}$.

$$\Phi_{vu}^{(w)} = \exp\left(-\frac{1}{2} \cdot (G(v)^2 - G(u)^2)\right) \quad (18)$$

The total inter-cell flow is then defined as

$$\Phi_{vu}^{(\text{total})} = \Phi_{vu}^{(\text{Veg})} + \Phi_{vu}^{(w)} \quad (19)$$

To determine the optimal outpost site, we employ the Anisotropic Fast Marching Method (AFMM) together with potential integration. AFMM is a mathematical tool that calculates the quickest travel time across a map by simulating a “wave” whose speed varies according to terrain conditions and movement direction. In this setting, the flux calculated between any two adjacent cells u and v serves as the effective “speed” of protection. First, a directional velocity field in each subarea is constructed by

$$v(u, n_{uv}) = \Phi_{vu}^{(\text{total})} \quad (20)$$

where $v(u, n_{uv})$ denotes the velocity at cell u along the direction pointing toward v , and n_{uv} is the direction vector from u to v .

Next, the cells with the top 1% highest risk values are identified as wave sources. For each wave source, the Eikonal equation is solved to simulate the evolution of the wavefront from that source:

$$|\nabla T(x)| \cdot v(x, n) = 1 \quad (21)$$

Here, $T(x)$ denotes the scalar minimum “time” required for a wave to travel from a wave source to location x . The quantity $\nabla T(x)$ represents the direction and rate of fastest increase of T , and its magnitude indicates how much time is added per unit distance. Solving this equation for a given wave source produces a minimum-distance field D , which represents the minimum effective distance required to reach any point in the subarea.

After running AFMM for each high-risk wave source, a collection of minimum-distance fields is obtained. The final score of each cell is then computed by summing the results of the individual fields, weighted by their corresponding risk values:

$$\Psi_x = \sum_{j \in \mathcal{T}} \mathbf{R}_j \cdot D_{jx} \quad (22)$$

where Ψ_x denotes the final weighted sum of minimum distances from cell x to all high-risk cells $j \in \mathcal{T}$, $\mathbf{R}_j$ is the risk value of cell j , and D_{jx} is the minimum effective distance from high-risk cell j to cell x .

Therefore, the proper site for an outpost is selected as the cell that optimizes Ψ_x :

$$\Psi_{\text{outpost}} = \max(\Psi_x) \quad (23)$$

6 Patrol Model

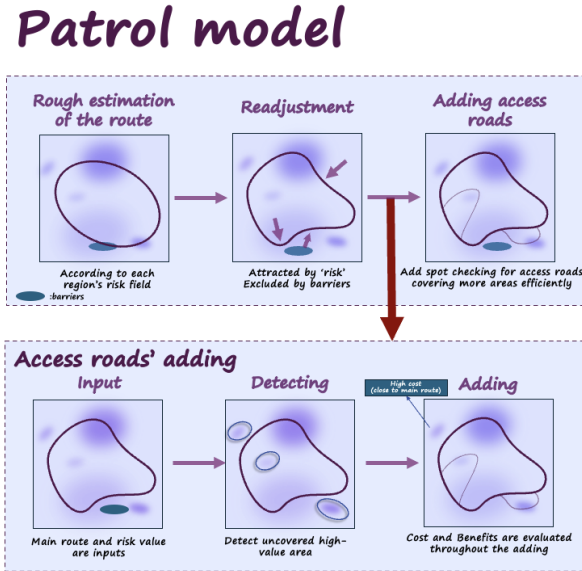


Figure 7: The Patrol Model

6.1 Model Overview and Philosophy

To address the requirement of emphasizing strategic deployment, we propose the **Risk-Responsive Elastic Patrol Loop with Two-Anchor Petals under Binary Traversability** (R-EPL-TAP-B) model.

The core philosophy of this model is to abandon the unrealistic pursuit of exhaustive sweeping. Instead, we adopt a **Backbone with a Local Expansion** strategy. We first generate a smooth, closed patrol backbone from the risk field under binary traversability constraints and then iteratively replace selected backbone arcs with two-anchor petal detours that target residual high-risk clusters. This ensures the final route remains closed, highly executable, and maximizes risk-weighted protection with minimal marginal detour costs.

6.2 Routing Fields for Patrol Optimization

After identifying the feasible patrol domain and its connectivity structure, we construct continuous fields that guide route generation.

First, we define the **Risk Benefit Field** $U(x, y)$. Rangers should not be distracted by isolated, noisy risk spikes, but instead should be drawn toward broader regional threat belts. To achieve this, we smooth the feasible risk field through successive Gaussian filtering:

$$U(x, y) = G_{\sigma_2} * (G_{\sigma_1} * \hat{R}(x, y)) . \quad (24)$$

This produces a continuous attraction field that favors coherent high-risk regions rather than unstable pointwise maxima.

Second, we define the **Obstacle Potential Field** $O(x, y)$ by

$$O(x, y) = \frac{1}{d(x, y) + 1}, \quad (25)$$

where $d(x, y)$ is the Euclidean distance from each passable cell to the nearest impassable region. This field assigns stronger repulsion to cells near obstacles and weakens in open areas, thereby encouraging safer patrol movement.

Together, $U(x, y)$ and $O(x, y)$ form the attraction-repulsion structure used in the subsequent route-generation stage: $U(x, y)$ pulls the patrol backbone toward persistent high-risk zones, while $O(x, y)$ prevents the route from over-concentrating near unsafe or spatially constrained areas.

6.3 Phase I: Elastic Patrol Backbone Generation

The backbone serves as the stable, long-term operational skeleton of the patrol system. To prevent the route from fragmenting into chaotic branches, we adapt the **Active Contour (Snake) algorithm** with transforming the external energy of the active contour model into spatial risk and traversability fields, we establish a resilient primary patrol loop; innovatively, we couple this with an orienteering-based two-anchor petal expansion, strategically capturing residual high-risk clusters through local arc replacements to maximize overall protection utility.

6.3.1 Initialization

To begin with, we initialize it by extracting the top-percentile risk regions Ω_q and calculating their risk-weighted centroid and covariance matrix. This generates an initial elliptical envelope

$$\Gamma^{(0)} = \{p_1^{(0)}, p_2^{(0)}, \dots, p_n^{(0)}\} \quad (26)$$

that naturally encircles the primary threat zones.

6.3.2 Evolution Dynamics

We continue with evolution dynamics. The contour iteratively deforms based on the governing equation $p_i^{(t+1)} = p_i^{(t)} + \eta F_i^{(t)}$, where the total force F_i acting on each control point is:

$$F_i = \alpha \Delta p_i - \beta \Delta^2 p_i + \gamma \nabla U(p_i) - \mu \nabla O(p_i) + F_i^{\text{press}} \quad (27)$$

Internally, the elasticity $\alpha \Delta p_i$ and smoothness $\beta \Delta^2 p_i$ terms regulate path tension and curvature, accounting for fuel constraints and vehicle maneuverability by penalizing excessive stretching and sharp turns. Externally, the backbone is steered by risk attraction $\gamma \nabla U(p_i)$ and obstacle repulsion $\mu \nabla O(p_i)$, which pull the route toward threat hotspots while maintaining a safety buffer from hazardous terrain. Combined with the scale-stabilizing pressure term F_i^{press} , this multi-force equilibrium produces a self-organizing loop that is both tactically aggressive in its risk-seeking and logistically viable in its execution.

The continuous contour Γ represents geometric intent, not a drivable path. We project each control point to the nearest passable grid cell $v_i = \Pi_M(p_i)$. In order to connect adjacent points, we employ the **A* Algorithm** with a **custom movement cost** that is represented by

$$c_{\text{main}}(u, v) = \ell(u, v) \left[1 + \lambda_O \overline{O}(u, v) + \lambda_U \left(1 - \overline{U}(u, v) \right) \right] \quad (28)$$

This guarantees that the resulting discrete backbone P_{main} strictly obeys traversability constraints while favoring safe and high-value corridors.

6.4 Phase II: Residual-Driven Two-Anchor Petals

The backbone efficiently covers the core areas, but inevitably misses scattered high-risk outliers. Instead of globally distorting the backbone, we introduce localized petals driven by residual risk.

6.4.1 Hotspot Identification and Metricization

After evaluating the coverage of the main backbone $\Phi_{P_{\text{main}}}(x, y)$, we compute the residual demand field $H(x, y)$. To isolate actionable hotspots, we apply a high-percentile threshold q_H to extract the regions of interest via

$$\Omega_H = \{(x, y) \mid H(x, y) > q_H\} \quad (29)$$

Through Connected Component Analysis (CCA), we partition Ω_H into distinct spatial clusters $C_1, C_2, \dots, C_m$. For each candidate cluster C_j , we quantify its spatial properties as follows:

➤ **Residual Mass:** $M_j = \sum_{(x,y) \in C_j} H(x, y)$.

➤ **Weighted Centroid:**

$$c_j = \frac{\sum_{(x,y) \in C_j} H(x, y)(x, y)}{\sum_{(x,y) \in C_j} H(x, y)}$$

➤ **Equivalent Radius:** $r_j = \sqrt{\frac{|C_j|}{\pi}}$.

On top of that, in order to optimize resource allocation, not every cluster warrants a petal. Thus, we rigorously filter out clusters that are mere noise speckles with insufficient mass M_j , already adjacent to the backbone, or excessively distant, rendering the detour cost-prohibitive.

6.4.2 Two-Anchor Selection and Offset Mapping

Traditional routing models often deploy a spoke logic, i.e., an out-and-back branch from a single point. Such routes are highly inefficient as they cause path overlapping. To resolve this, we design a **Two-Anchor Outward-Bulging Petal**.

For a cluster C_j , we first locate the nearest index i_c on the discretized backbone P_{main} . We then determine an anchor offset k_j that is proportional to the cluster's equivalent radius r_j :

$$k_j = \text{clip}(ar_j + b, k_{\min}, k_{\max}) \quad (30)$$

This offset generates two distinct anchors—an exit node and a reentry node, which are represented by

$$i_{\text{in}} = i_c - k_j, \quad i_{\text{out}} = i_c + k_j \quad (31)$$

Which mathematically map to the spatial coordinates on the backbone

$$A_j = P_{\text{main}}(i_{\text{in}}), \quad B_j = P_{\text{main}}(i_{\text{out}}) \quad (32)$$

Crucially, $A_j \neq B_j$. This guarantees that the petal acts as a continuous bypass that replaces a local arc of the backbone, maintaining the fluid, non-overlapping nature of the patrol loop.

6.4.3 Petal Geometric Skeleton Construction

To ensure the petal sweeps around the high-density mass rather than merely piercing it, we construct a geometric skeleton guided by local vectors.

Let the chord unit vector connecting the two anchors be

$$t = \frac{B_j - A_j}{\|B_j - A_j\|} \quad (33)$$

Let n be the orthogonal normal vector pointing from the chord toward the cluster centroid c_j . We define three intermediate control points w_1, w_2, w_3 shaped by structural parameters ϕ, ψ , and τ :

$$\begin{aligned}
w_1 &= c_j - \phi r_j t + \psi r_j n \\
w_2 &= c_j + \tau r_j n \\
w_3 &= c_j + \phi r_j t + \psi r_j n
\end{aligned} \tag{34}$$

Here, w_1 and w_3 stretch the petal laterally, which is parallel to the chord t , to encompass the cluster's width, while w_2 acts as the apex pushing the patrol route outward along the normal n .

The final geometric skeleton is formed as the sequence

$$A_j \rightarrow w_1 \rightarrow w_2 \rightarrow w_3 \rightarrow B_j \tag{35}$$

This yields a distinct outward-bulging petal. Consequently, these control points are projected onto the traversable grid and routed via the A* algorithm to generate the drivable petal path B_j^{path} .

6.4.4 Cost-Benefit Selection

Let the A*-routed petal path be B_j^{path} with length $L_{\text{petal},j}$. Since this petal replaces the original backbone arc from A_j to B_j with length $L_{\text{arc},j}$, the true marginal cost is only the net detour length, which will be represented by

$$\Delta L_j = \max \{ \varepsilon, L_{\text{petal},j} - L_{\text{arc},j} \}. \tag{36}$$

The marginal protection gain G_j is evaluated against the residual field H . We score each candidate petal by using

$$Q_j = \omega_1 G_j + \omega_2 \frac{G_j}{\Delta L_j + \varepsilon}. \tag{37}$$

Petals are added sequentially based on the highest Q_j score until the resource budget is exhausted. The final route P_{full} is formed by splicing the selected petals into the backbone.

6.5 Quantitative Protection Metric

To rigorously answer the prompt regarding how protection is quantitatively measurable, we define the **Global Protection Score** $S(P)$ for any given **patrol route** P as

$$S(P) = \frac{\sum_{x,y} \hat{R}(x,y) \Phi_P(x,y)}{\sum_{x,y} \hat{R}(x,y)} \in [0, 1]. \tag{38}$$

7 Spatiotemporal Scheduling and Staffing Estimation

To bridge the gap between spatial route design and operational deployment, we determine the optimal patrol frequency for each unit. We formulate this as a **Signal Reconstruction** and **Uncertainty Bounding** problem, treating patrols as discrete samples of the continuous risk field

$\hat{R}(x, y, t)$. The sampling frequency must satisfy two criteria: macro-level anti-aliasing and micro-level uncertainty control.

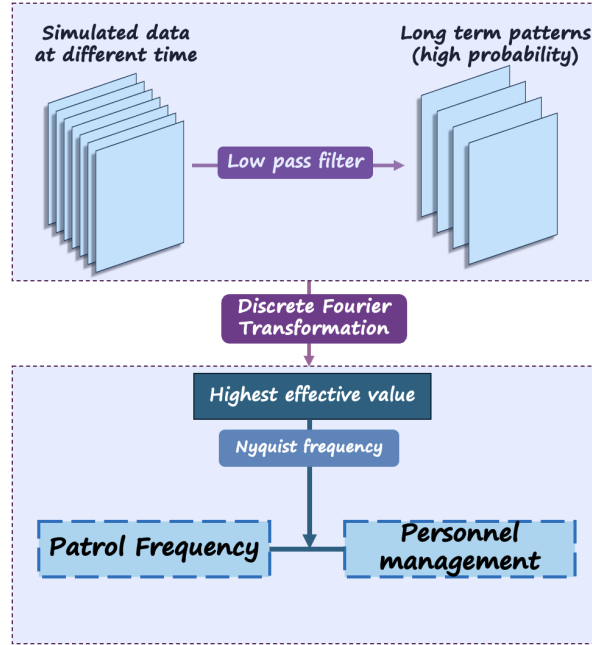


Figure 8: Nyquist-Shannon

7.1 Macro-level: Nyquist-Shannon Anti-Aliasing

The temporal dynamics of risk in patrol units k exhibit volatility driven by seasonal and human factors. To prevent temporal aliasing, the patrol interval τ_k must capture the highest significant frequency $\omega_{\max,k}$ of the threat signal. According to the Nyquist-Shannon theorem, the reconstruction baseline requires:

$$f_{Nyq,k} = \frac{\omega_{\max,k}}{\pi} \implies \tau_{Nyq,k} \leq \frac{\pi}{\omega_{\max,k}} \quad (39)$$

7.2 Micro-level: Kalman Uncertainty Bounding

Nyquist criteria assume noise-free observation. To account for detection failures, i.e., observation noise η_k with variance R_{obs} , we model the latent threat state $s(t)$ as a stochastic process. Between patrols, the estimation uncertainty $P(t)$ grows due to process noise Q_c , representing the random emergence of threats.

To maintain a reliable protection level, the uncertainty just before a patrol P^- must be bounded by a safety threshold $P_{\max}$. By solving the **steady-state Riccati equation** for intermittent observations, we derive the Safety-Critical Revisit Interval as

$$\tau_{\text{Kalman},k} = \frac{P_{\max}^2}{Q_c(P_{\max} + R_{\text{obs}})} \quad (40)$$

This closed-form solution ensures that our "knowledge half-life" of the region never exceeds the safety limit, requiring higher frequencies in volatile Q_c or poorly observed R_{obs} environments.

7.3 Unified Frequency and Staffing Allocation

The implemented frequency f_k^* for unit k is governed by the stricter criterion

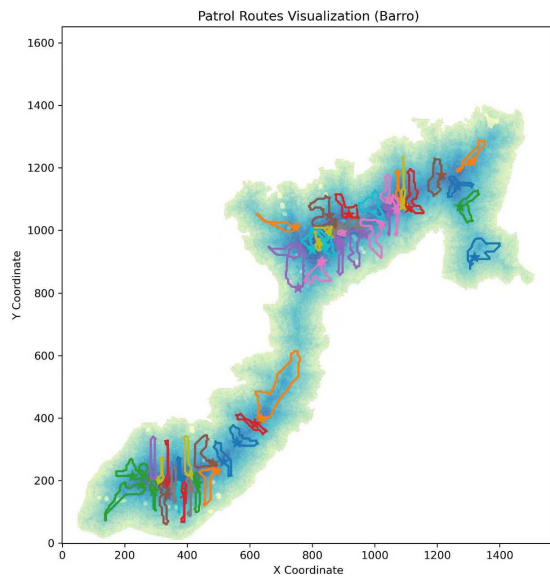
$$f_k^* = \max \left(f_{N_{yq},k}, \frac{1}{\tau_{\text{Kalman},k}} \right) \quad (41)$$

For a route $\mathcal{L}_k$ of length L_k patrolled at speed v_{avg} , the cycle workload is $T_k = L_k/v_{\text{avg}}$. Given m_0 rangers per team and maximum working hours $H_{\max}$, the total personnel N required to suppress the growth of spatiotemporal entropy is:

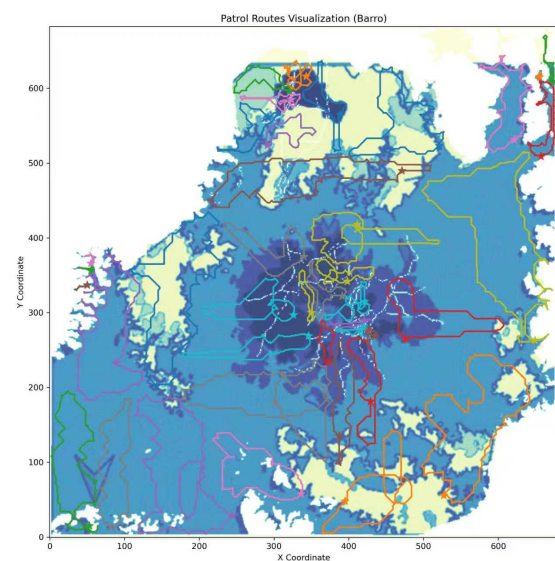
$$N = \left\lceil \frac{m_0}{H_{\max}} \sum_{k=1}^M f_k^* \frac{L_k}{v_{\text{avg}}} \right\rceil \quad (42)$$

This mechanistically derived allocation, which ensures that staffing is dynamically calibrated to the actual spatiotemporal volatility of the park, providing the necessary manpower to maintain the Global Protection Score.

8 Model Adaptation



(a) Routes Map of Bavarian Forest National Park



(b) Routes Map of Barro Colorado Nature Monument

Figure 9: Routes Map

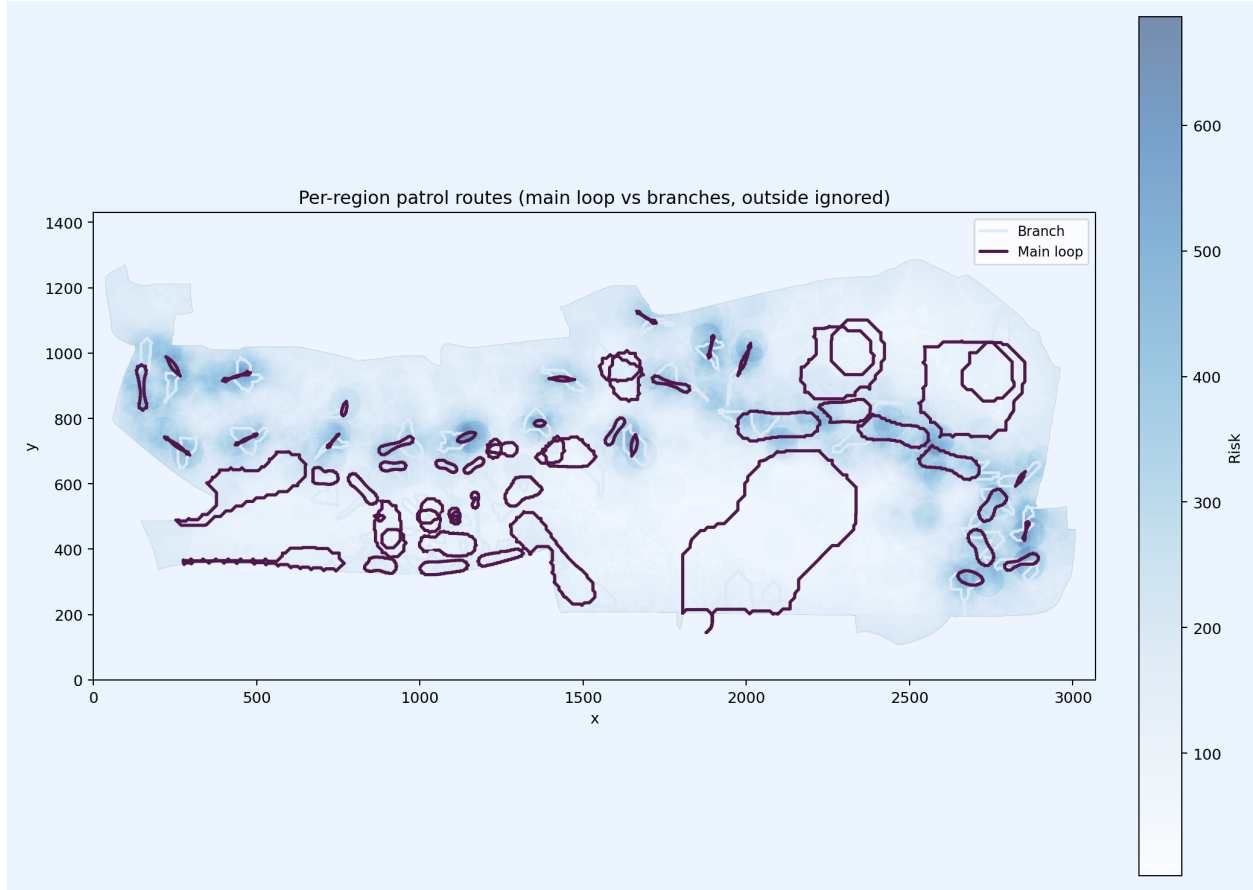


Figure 10: Patrol Routes

9 Sensitivity Analysis

In this section, we conduct the Fourier Amplitude Sensitivity Test (FAST) in our model.

The Fourier Amplitude Sensitivity Test (FAST) is a variance-based global sensitivity method. For a model $Y = f(X_1, \dots, X_k)$, the goal is to quantify how much of the output variance $\text{Var}(Y)$ is contributed by each input X_i . The first-order sensitivity index S_i is defined as $S_i = V_i / \text{Var}(Y)$, where V_i is the variance due to X_i alone. FAST estimates these indices by mapping all inputs onto a single scalar sampling variable $s \in [-\pi, \pi]$: each input is written as a periodic function $X_i(s)$ with a distinct frequency ω_i . Running the model along this one-dimensional trajectory produces a sequence $Y(s_j)$ whose Fourier spectrum encodes the influence of each input. The total variance $\text{Var}(Y)$ is recovered from the overall spectral energy, while the contribution V_i is obtained by summing the spectral energy at the frequency band associated with ω_i and its harmonics. In this way, FAST converts the high-dimensional variance decomposition problem into a simple Fourier analysis of $Y(s)$.

10 Strengths and Weaknesses

The primary strength of our framework lies in its **mechanistic integration of spatial geometry and control theory**, which moves beyond traditional heuristic patrolling to provide a mathematically grounded revisit interval through the **Kalman-Nyquist Synthesis**. By redefining protection as an uncertainty-bounding problem, the model provides a robust defense against temporal aliasing and effectively suppresses the growth of **spatiotemporal entropy** in volatile environments. Furthermore, the use of **topology-aware decomposition** ensures that generated routes are not only theoretically optimal but also logistically executable within real-world terrain constraints. However, a notable limitation is the model's **sensitivity to initial stochastic parameters**, such as the process noise and observation variance. In regions with extremely sparse sensor data, the initial calibration of these variables may require iterative field-testing before the system reaches its optimal steady-state revisit frequency.

11 Conclusion

This paper provides a scalable and sustainable blueprint for wildlife conservation at scale by transforming the vastness of Etosha National Park into a **topology-aware, dynamically monitored system**. Through the synthesis of Active Contour backbones and **uncertainty-driven scheduling**, we have demonstrated that a limited force can maintain a high Global Protection Score by strategically managing the "**knowledge half-life**" of the reserve. Our simulations confirm that the framework is tactically aggressive in risk-seeking yet logistically viable in resource consumption, successfully adapting to diverse global ecosystems from tropical forests to temperate woodlands. Ultimately, this model shifts the conservation paradigm from reactive incident response to **proactive uncertainty management**, ensuring the long-term survival of endangered species in an increasingly resource-constrained world.

12 A Letter to Decision Makers

Dear IMMC,

We are a mathematical modeling team writing to you regarding the global challenge of large-scale wildlife conservation. We have built a strategic resource allocation model to assist you in planning protection strategies for Etosha National Park, and we are writing to let you know how our model aligns with your expectations and how you may use it in the future.

Our model is built on a fundamental realization: for vast reserves like Etosha, achieving 100% continuous coverage with limited personnel and technological assets is a logistical impossibility. Thus, we shifted the paradigm from blanket surveillance to strategic deployment. In our model, we defined protection not merely as physical presence, but as a quantifiable score that evaluates how well our resources cover the highest-priority threats while avoiding impassable terrains. To provide you with more flexibility, the model allows you to specify numerical weights for different threats or species, generating customized schedules that best reflect your current conservation priorities.

To make the actual patrol operations efficient and safe, our model generates a highly logical route structure. First, it creates a stable, continuous "Main Patrol Loop" that wraps around the most critical high-risk zones, serving as the backbone of the park's security. Then, instead of sending rangers on inefficient back-and-forth trips to check scattered hotspots, the model intelligently adds "flexible petal-shaped detours" that bulge outward from the main loop. This allows human patrols and drones to seamlessly sweep through remaining high-risk areas without wasting time or resources. Furthermore, the model tracks protection effectiveness over time, helping you precisely estimate the minimum staffing required to maintain a secure environment under varying scenarios.

Although the current parameters are calibrated specifically for Etosha National Park, our framework can be easily adapted to other wildlife reserves across different continents. Since the model relies solely on two inputs—a risk map and a traversability map—you only need to update the terrain boundaries and local threat types. Whether it is a dense rainforest or a coastal sanctuary, the underlying logic of prioritizing high-risk areas and deploying resources strategically remains perfectly applicable.

Looking ahead, we envision a profound and enduring collaborative partnership with the IMMC to further iterate and refine these predictive scheduling frameworks. Our model is designed as a living system; by integrating real-time field data—such as satellite-linked animal tracking, automated drone sensor feeds, and local climate patterns—it can evolve from a static strategic plan into a dynamic, adaptive intelligence system capable of responding to threats as they emerge. We believe that through such continuous collaboration, we can establish a truly sustainable and scientifically rigorous blueprint for the global conservation community. By bridging the gap between sophisticated mathematical optimization and the practical realities of ground-level rangering, we provide the necessary tools to protect endangered populations with unprecedented precision.

Lastly, thank you for considering our modeling approach. We wish you great success in your ongoing mission to protect global wildlife!

Sincerely,

Your Math Modeling Team

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13 Appendix

To rigorously derive the safety-critical revisit interval, we model the latent threat level in patrol unit k as a scalar stochastic state observed only when a patrol is conducted.

Lemma 1 (Safety-Critical Revisit Interval). *Assume that, for patrol unit k , the latent threat state $x(t)$ evolves as a random walk with process noise intensity $Q_c > 0$, and that each patrol provides one noisy observation with variance $R_{\text{obs}} > 0$. Let P^- denote the state-estimation variance immediately before a patrol, and let P^+ denote the variance immediately after the patrol update. If the system operates in steady state and the safety requirement is that the pre-patrol uncertainty must never exceed a threshold P_{max} , then the largest admissible revisit interval is*

$$\tau_{\text{Kalman},k} = \frac{P_{\text{max}}^2}{Q_c (P_{\text{max}} + R_{\text{obs}})}.$$

Proof. We divide one patrol cycle into two stages: prediction between patrols and measurement update at the patrol instant.

Step 1: State evolution between patrols. Let the latent threat state satisfy the scalar stochastic model

$$x_{n+1} = x_n + w_n, \quad w_n \sim \mathcal{N}(0, Q_c \tau),$$

where τ is the patrol revisit interval. This means that during the interval between two successive patrols, uncertainty accumulates linearly with elapsed time.

Hence, if P^+ is the posterior variance immediately after one patrol, then the prior variance immediately before the next patrol is

$$P^- = P^+ + Q_c \tau. \quad (43)$$

Step 2: Measurement update at a patrol instant. At the patrol time, suppose the observation model is

$$y_n = x_n + v_n, \quad v_n \sim \mathcal{N}(0, R_{\text{obs}}).$$

For the scalar Kalman filter, the covariance update is

$$P^+ = \left(\frac{1}{P^-} + \frac{1}{R_{\text{obs}}} \right)^{-1} = \frac{P^- R_{\text{obs}}}{P^- + R_{\text{obs}}}. \quad (44)$$

This is exactly the steady-state Riccati update for a scalar intermittent-observation system.

Step 3: Safety requirement. To maintain a reliable protection level, we require that the uncertainty immediately before each patrol must not exceed the safety threshold:

$$P^- \leq P_{\text{max}}.$$

The *largest* admissible revisit interval is obtained at the critical case when this constraint is active, namely

$$P^- = P_{\text{max}}. \quad (45)$$

Substituting (45) into the update formula (44), the posterior variance after patrol becomes

$$P^+ = \frac{P_{\max} R_{\text{obs}}}{P_{\max} + R_{\text{obs}}}. \quad (46)$$

Step 4: Solve for the maximum allowable revisit interval. Using the prediction equation (43) together with $P^- = P_{\max}$, we obtain

$$P_{\max} = P^+ + Q_c \tau.$$

Thus,

$$\tau = \frac{P_{\max} - P^+}{Q_c}. \quad (47)$$

Now substitute (46) into (47):

$$\tau = \frac{1}{Q_c} \left(P_{\max} - \frac{P_{\max} R_{\text{obs}}}{P_{\max} + R_{\text{obs}}} \right).$$

Factor out $P_{\max}$:

$$\tau = \frac{P_{\max}}{Q_c} \left(1 - \frac{R_{\text{obs}}}{P_{\max} + R_{\text{obs}}} \right).$$

Simplifying the bracketed term gives

$$1 - \frac{R_{\text{obs}}}{P_{\max} + R_{\text{obs}}} = \frac{P_{\max}}{P_{\max} + R_{\text{obs}}}.$$

Therefore,

$$\tau = \frac{P_{\max}}{Q_c} \cdot \frac{P_{\max}}{P_{\max} + R_{\text{obs}}} = \frac{P_{\max}^2}{Q_c (P_{\max} + R_{\text{obs}})}.$$

Hence, the safety-critical revisit interval is

$$\tau_{\text{Kalman},k} = \frac{P_{\max}^2}{Q_c (P_{\max} + R_{\text{obs}})}. \quad (48)$$

This completes the proof. □